Università degli Studi di Milano

Statistical Methods for Machine Learning — A.Y. 2025/26

Experimental Project, Assignment 1

Adaptive Hedge vs. Hedge with Fixed Learning Rate

An experimental study of prediction with expert advice

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Code repository: `https://github.com/Nitinydv69/Adaptive_hedge-ML`

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Declaration

I declare that this material, which I now submit for assessment, is entirely my own work and has not been taken from the work of others, save and to the extent that such work has been cited and acknowledged within the text of my work. I understand that plagiarism, collusion, and copying are grave and serious offences in the university and accept the penalties that would be imposed should I engage in plagiarism, collusion or copying. This assignment, or any part of it, has not been previously submitted by me or any other person for assessment on this or any other course of study.

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Abstract

I implement from scratch the Hedge algorithm with a fixed learning rate η and the parameter-free AdaHedge algorithm [1, 2], and compare them on four synthetic loss sequences ($K = 10$, $T = 5000$, 20 seeds): a stochastic sequence with a large gap, one with a tiny gap, a piecewise-stationary ("switching") sequence, and a deterministic sequence that is a worst case for follow-the-leader. No fixed η is good everywhere: $\eta = 0.5$ gives regret ≈ 7 on the stochastic sequence but ≈ 310 (linear in T) on the alternating one, while the worst-case-optimal η^* is safe but $7\times$ worse than AdaHedge on the stochastic sequence (39.9 vs. 5.5). AdaHedge is better than every realisable fixed choice I tried on all three informative sequences, and within about 1.3 regret units of a hindsight-tuned oracle (a small but statistically detectable gap) on the two stochastic-type ones. A scaling experiment ($T = 500$ to $50\,000$) shows its regret flat on easy data and growing like $\sqrt{T}$ on the hard sequence. On the nearly indistinguishable low-gap sequence no algorithm is distinguishable from another, and I explain why. I trace the behaviour to $\eta_t = \ln K/\Delta_{t-1}$ and verify numerically that $R_T \leq 2\Delta_T$.

1 Introduction

Motivation. Online learning gives guarantees that hold for *every* sequence, without probabilistic assumptions. In the course this is developed through online gradient descent over continuous decision sets; prediction with expert advice is the combinatorial counterpart: a learner combines the advice of K experts, and the standard algorithm, HEDGE, depends on a learning rate η. Too large an η makes the algorithm chase noise; too small makes it slow. The theory-optimal value needs the horizon T and is tuned for the *worst* case, which can be very pessimistic on easy data. The paper *Adaptive Hedge* [1] removes this tuning.

Objectives. I (i) implement HEDGE and ADAHEDGE from scratch, (ii) compare them with several η on stochastic, low-gap and switching sequences, (iii) identify the regimes where adaptivity helps, and (iv) connect the behaviour to the theory. Beyond the minimum I add: an η-sweep with the theoretical bound overlaid; the trajectory η_t of ADAHEDGE; a deterministic worst-case sequence; a scaling study in T; paired confidence intervals; and a numerical check of $R_T \le 2\Delta_T$.

2 Background

2.1 Problem setting

At each round $t = 1, \dots, T$ the learner picks $w_t \in \Delta_K$ (the probability simplex), then observes $\ell_t \in [0,1]^K$ and suffers $h_t = \langle w_t, \ell_t \rangle$. The regret against the best fixed expert is

$$R_T = \sum_{t=1}^{T} h_t - \min_k L_{T,k}, \qquad L_{T,k} = \sum_{t=1}^{T} \ell_{t,k}.$$

2.2 Hedge

HEDGE plays $w_{t,k} \propto \exp(-\eta L_{t-1,k})$ (Algorithm 1). For every sequence, $R_T \le \frac{\ln K}{\eta} + \frac{\eta T}{8}$, which is minimised by $\eta^* = \sqrt{8 \ln K / T}$ and gives $R_T \le \sqrt{(T/2) \ln K}$. This needs T in advance, and it is a worst-case statement, pessimistic on easy data.

2.3 AdaHedge

Define the *mix loss* $m_t = -\frac{1}{\eta_t} \ln \sum_k w_{t,k} e^{-\eta_t \ell_{t,k}}$ and the *mixability gap* $\delta_t = h_t - m_t \ge 0$; let $\Delta_t = \sum_{s \le t} \delta_s$. ADAHEDGE uses $\eta_t = \ln K / \Delta_{t-1}$, with $\eta_1 = \infty$ (Algorithm 2). The gap δ_t measures how much the learner loses relative to an idealised mix prediction; it is small when the weights are concentrated (easy data) and large when experts disagree and the leader keeps changing. The regret is controlled by this data-dependent quantity: with $M_T = \sum_t m_t$ we have $R_T = \Delta_T + (M_T - L_T^*)$, and for non-increasing learning rates $M_T - L_T^* \le \ln K / \eta_T = \Delta_{T-1}$, hence

$$R_T \le 2\Delta_T \qquad \text{[2] (proof sketch above)}, \tag{1}$$

and $\Delta_T = O(\sqrt{T \ln K})$ in the worst case [1, 2] (quoted up to constants and lower-order terms). So ADAHEDGE keeps a worst-case guarantee of the same order as HEDGE(η^*) without tuning, and can be far better when Δ_T is small.

Algorithm 1 HEDGE with fixed η

$L \leftarrow 0 \in \mathbb{R}^K$
for $t = 1, \dots, T$ **do**
  $w_k \propto \exp(-\eta(L_k - \min_j L_j))$
  play w; observe ℓ_t; suffer $h_t = \langle w, \ell_t \rangle$; $L \leftarrow L + \ell_t$

Algorithm 2 AdaHedge

$L \leftarrow 0$, $\Delta \leftarrow 0$
for $t = 1, \dots, T$ **do**
  if $\Delta = 0$ **then** $\eta \leftarrow \infty$; $w \leftarrow$ uniform on $\arg\min_k L_k$
  else $\eta \leftarrow \ln K / \Delta$; $w_k \propto \exp(-\eta(L_k - \min_j L_j))$
  play w; observe ℓ_t; $h \leftarrow \langle w, \ell_t \rangle$
  $m \leftarrow \min_{k:w_k>0} \ell_{t,k}$ if $\eta = \infty$, else $-\frac{1}{\eta} \ln \sum_k w_k e^{-\eta \ell_{t,k}}$
  $\Delta \leftarrow \Delta + \max(0, h - m)$; $L \leftarrow L + \ell_t$

3 Experimental methodology

3.1 Implementation

Both algorithms use NumPy only (`src/algorithms.py`). The exponent is shifted by the minimum cumulative loss for numerical stability. In AdaHedge the mix loss is computed with a shift by the minimum loss over the *support* of w_t, which is exact for $\eta = \infty$ and avoids underflow for large η. (My first version shifted by the minimum over all experts, which gave $\delta_t \approx 1$ instead of 0 in the $\eta = \infty$ phase; I found this with a two-round counterexample and fixed it before the experiments below.)

3.2 Loss sequences

All sequences use $K = 10$, $T = 5000$ and binary losses (except $t = 1$ in the last one). They differ in how hard it is to identify and follow the best expert.

Stochastic. Expert 1 $\sim$ Bernoulli(0.3), the other nine Bernoulli(0.5) (gap 0.2).

Low gap. Best expert Bernoulli(0.49), the others 0.5 (gap 0.01, below the statistical resolution $\approx 1/\sqrt{T} \approx 0.014$).

Adversarial (switching). Three equal segments; in segment s expert s is Bernoulli(0.2), the others 0.5. The best expert changes over time, but the sequence is random and oblivious, so it is "adversarial" only in that sense. The best *fixed* expert has mean loss ≈ 0.4.

Alternating (extra). Deterministic. Experts 1 and 2 alternate between losses 0 and 1 in opposite phase; the other experts always lose 1. The leader flips every round, so a greedy learner loses ≈ 1 per round while the best expert loses $\approx T/2$. This is a worst case for follow-the-leader and for large η; Section 4.6 shows that fixed-η Hedge is also close to its own bound here.

3.3 Protocol and metrics

20 seeds $(0, \dots, 19)$ per setting; all algorithms see the *same* sequence for a given seed (paired comparison). Hedge is run with $\eta \in \{0.005,\ \eta^* \approx 0.061,\ 0.5\}$ and on a grid of 11 log-spaced values in $[10^{-2.5}, 10^{0.5}]$; the best grid value per setting serves as a hindsight *oracle* reference (not a realisable algorithm, and optimistic, since it is selected on the same seeds it is evaluated on). Metrics: regret at time t against the best fixed expert up to t, final regret (mean $\pm$ std), cumulative loss, and for AdaHedge the trajectory η_t and final gap Δ_T. Confidence intervals are 95% t-intervals over seeds (19 degrees of freedom).

4 Results

4.1 Overview

Figure 1 shows regret over time and Table 1 the final regrets.

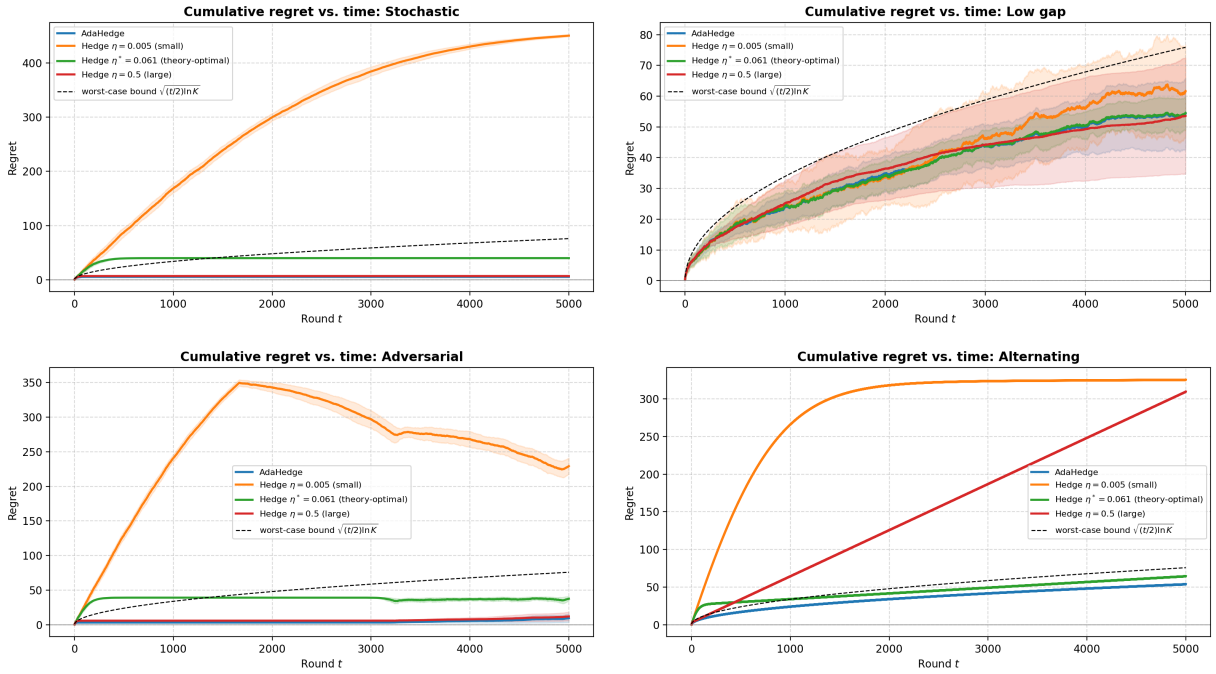


Figure 1: Cumulative regret vs. time (mean ± std, 20 runs). Dashed: $\sqrt{(t/2)\ln K}$, the HEDGE worst-case bound when tuned for horizon t.

	Stochastic	Low gap	Switching	Alternating
ADAHEDGE (no tuning)	5.5 ± 1.8	54.3 ± 11.5	9.6 ± 6.0	54.1 ± 0.0
HEDGE $\eta = 0.005$	450.2 ± 1.8	61.7 ± 15.0	229.1 ± 11.9	325.3 ± 0.0
HEDGE $\eta^* = 0.061$	39.9 ± 0.2	54.6 ± 5.4	37.7 ± 3.4	64.7 ± 0.0
HEDGE $\eta = 0.5$	6.9 ± 0.6	53.6 ± 18.8	12.1 ± 6.2	309.6 ± 0.0
HEDGE, best grid η (oracle)	4.2 ± 1.7	52.2 ± 13.9	8.3 ± 6.5	63.7 ± 0.0

Table 1: Regret at $T = 5000$, mean ± std over 20 runs. The oracle row uses hindsight to pick η and is not realisable. The alternating sequence is deterministic, hence std = 0.

4.2 Stochastic (large gap)

HEDGE with $\eta = 0.005$ stays near-uniform for thousands of rounds: its regret is still growing at T (450). With a fixed η the regret on easy data saturates at a level of order $\ln K/\eta$ (the price of learning which expert is best): for η^* it flattens at ≈ 40 after ≈ 500 rounds (compare $\ln K/\eta^* \approx 38$). With $\eta = 0.5$ and ADAHEDGE the regret is a small constant (6.9 and 5.5). ADAHEDGE is about $7\times$ better than η^*. *This is the regime where adaptivity helps most.*

4.3 Low gap

All methods finish at 52–62 and the differences are smaller than the spread over seeds, so *no algorithm can be declared better here.* The reason is not the algorithms: the regret is dominated by the comparator. Over the 20 runs the best expert beats the average loss $T/2$ by 69.0 (pure noise would give $\approx 1.54 \cdot \sqrt{T}/2 \approx 54$ from the minimum of ten Bernoulli(0.5) sums, the rest coming from the true 0.01 gap), while ADAHEDGE itself ends 14.7 *below* $T/2$. The comparator is "lucky" by an amount no online algorithm can match. To check this directly, Table 2 reports the *pseudo-regret* against expert 1, the best in expectation: all learners sit at ≈ 30, only modestly below uniform weights (39), and indistinguishable from each other. The ≈ 23 difference to the realised regret (54) is exactly how much the best realised expert beats expert 1. So at this gap and horizon, the sequence simply does not contain enough information to identify the best expert, whichever

Algorithm	Pseudo-regret vs. expert 1 (mean ± 95% CI)
AdaHedge	31.0 ± 13.5
Hedge η^*	31.2 ± 12.9
Hedge $\eta = 0.5$	30.2 ± 13.6
Uniform weights (no learning)	39.0 ± 16.4

Table 2: Low gap, $T = 5000$: $\sum_t h_t - L_{T,1}$, i.e. regret against the best expert *in expectation* rather than in hindsight.

comparator is used.

4.4 Switching best expert

AdaHedge (9.6) and $\eta = 0.5$ (12.1) clearly beat η^* (37.7) and $\eta = 0.005$ (229). Large η is not punished because losses are random with large within-segment gaps. Regret is measured against the best *fixed* expert, whose loss rate (≈ 0.4) is worse than the segment-best expert (0.2). Hedge and AdaHedge cannot exploit this gap: their weights depend on *cumulative* losses, so they never forget earlier segments and only converge to a good fixed expert; tracking the switching comparator would need a variant such as Fixed Share. The sequence is therefore a weak stress test of adaptivity, which motivates the alternating one.

4.5 Alternating (deterministic worst case)

Large η fails here: $\eta = 0.5$ follows the flipping leader and its regret is 309.6, close to near-uniform $\eta = 0.005$ (325). η^* (64.7) and AdaHedge (54.1) stay below the worst-case bound (≈ 76). AdaHedge is below every grid value, including the oracle (63.7); with a single deterministic sequence there is no seed-to-seed spread and my η grid is coarse, so I do not claim this is significant. The key finding is qualitative: *no single η is good both here and on the stochastic sequence*, while AdaHedge is near-best in both.

4.6 Hedge vs. its theoretical bound

Figure 2 plots final regret against η together with the bound $\ln K/\eta + \eta T/8$. On the alternating sequence the observed curve follows the bound closely on both sides of the minimum (within a factor ≈ 2 over the whole range), so this sequence is nearly a worst case for *every* fixed η, and the best fixed η is $\approx \eta^*$. On the stochastic and switching sequences the large-η branch of the bound is very loose: the observed regret keeps decreasing as η grows, and the best fixed η is at the edge of the grid. Thus the minimiser moves from $\eta \approx \eta^*$ to very large η, and AdaHedge (red line) tracks whichever is right. On low gap the curve is flat: when data is hard to read, η barely matters.

4.7 Scaling with the horizon

Figure 3 varies $T \in \{500, 5000, 50\,000\}$ for the stochastic (20 seeds; 10 at $T = 50\,000$ for runtime) and alternating (single deterministic sequence) settings, with Hedge at $\eta = 0.5$ and at the horizon-tuned $\eta^*(T)$. On the stochastic sequence AdaHedge stays flat (6.2, 5.5, 5.2) as does $\eta = 0.5$ (7.2, 6.9, 7.0), whereas $\eta^*(T)$ grows like $\sqrt{T}$ (14.3, 39.9, 122.0): the worst-case tuning pays a growing price on easy data. On the alternating sequence, $\eta = 0.5$ grows linearly (34.0, 309.6, 3065; a factor ≈ 10 per decade), while AdaHedge (17.4, 54.1, 170.1) grows by $\approx 3.1 \approx \sqrt{10}$ per decade, i.e. like $\sqrt{T}$, as does $\eta^*(T)$ (20.6, 64.7, 204.1). This supports the claim that AdaHedge behaves like a horizon-free η^* on hard data and like a large η on easy data.

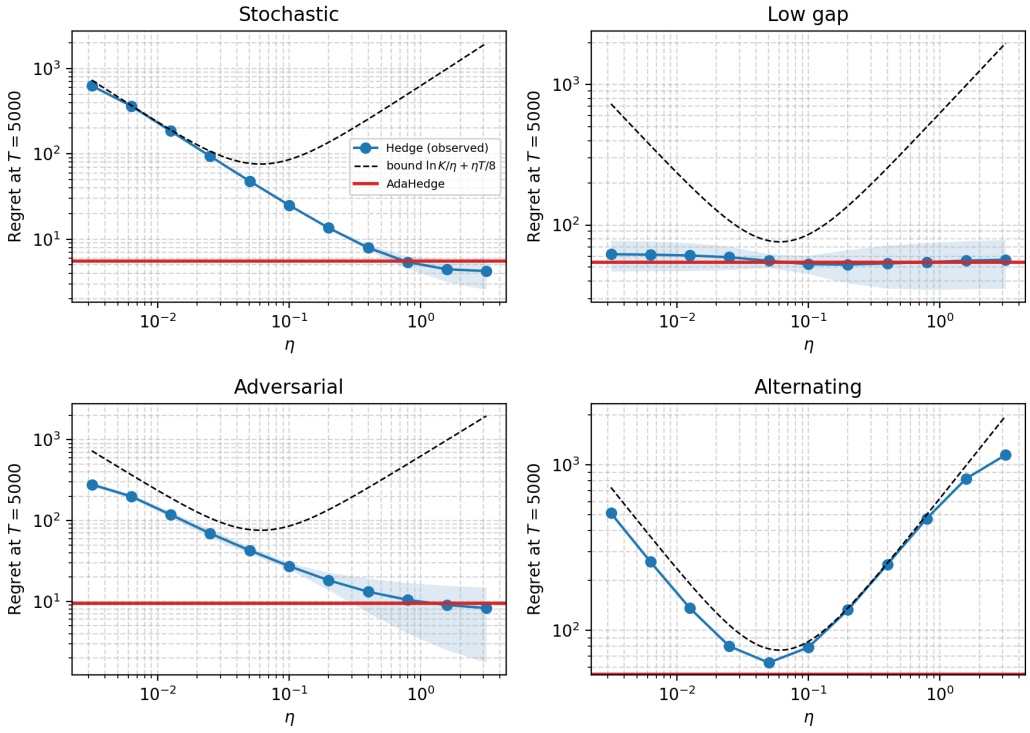


Figure 2: Final regret of HEDGE vs. η (blue, mean $\pm$ std, log–log) with the bound $\ln K/\eta + \eta T/8$ (dashed); ADAHEDGE needs no tuning (red).

4.8 Learning-rate behaviour

Figure 4 shows why ADAHEDGE adapts. On the stochastic sequence η_t stops decreasing after ≈ 60 rounds at ≈ 0.66 ($\Delta \approx 3.5$): once the weights concentrate on the best expert, $\delta_t \to 0$, Δ_t stops growing, and the large learning rate is retained, giving constant regret. On the hard sequences the weights never concentrate, δ_t keeps contributing, and η_t decays like $1/\sqrt{t}$: a least-squares fit of the log–log median curve on $t \geq 500$ gives slope -0.47 (low gap) and -0.50 (alternating), recovering the scaling $\eta^* \propto 1/\sqrt{T}$. On the switching sequence the slope is -0.26: η_t sits on a plateau (≈ 1) until $t \approx 3300$, about the start of the third segment, and only then decreases; on the stochastic one the slope is 0.00.

4.9 Cumulative loss

The assignment asks for cumulative-loss plots; for all four settings they are in Appendix A. Because all curves are $\approx t/2$ they look identical at full scale, so Figure 5 adds a zoom on the last 500 rounds for two settings, where the separations of Table 1 become visible: on the alternating sequence $\eta = 0.5$ is far above the others, and on the stochastic sequence η^* lags while ADAHEDGE and $\eta = 0.5$ track the best expert.

4.10 Statistical comparison

Table 3 gives paired differences (ADAHEDGE minus the reference, per seed, mean $\pm$ 95% CI); negative means ADAHEDGE is better. ADAHEDGE is significantly better than η^* on the stochastic and switching sequences (-34.4, -28.1) and than $\eta = 0.5$ (-1.3 ± 0.6, -2.5 ± 0.7). Against the hindsight oracle it is slightly but detectably *worse* ($+1.3 \pm 0.4$ and $+1.3 \pm 0.9$), so a "within one standard deviation" reading of Table 1 would understate this; since the oracle is selected on the same seeds and sits at the edge of the grid, this comparison favours the oracle. On low gap every interval contains zero. On the alternating sequence there is a single deterministic sequence, so I

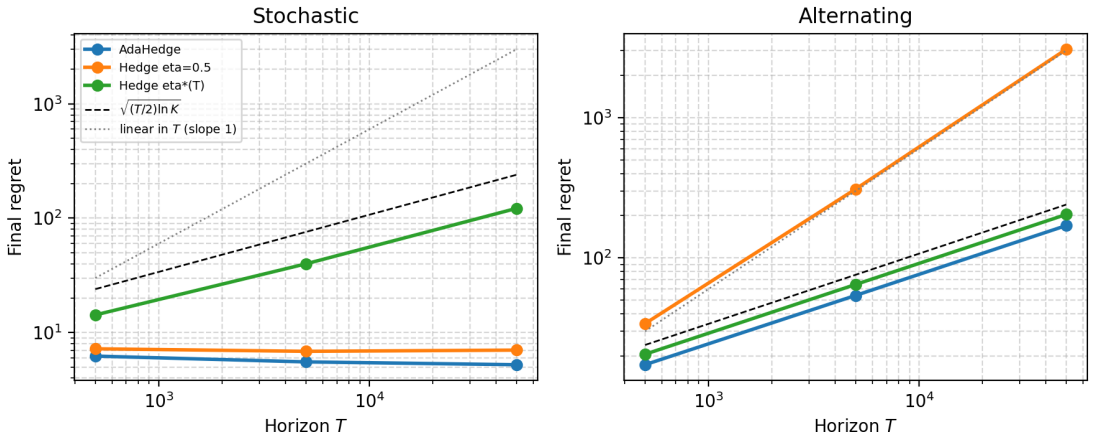


Figure 3: Final regret vs. horizon T (log–log). Dashed: $\sqrt{(T/2)\ln K}$; dotted: linear growth.

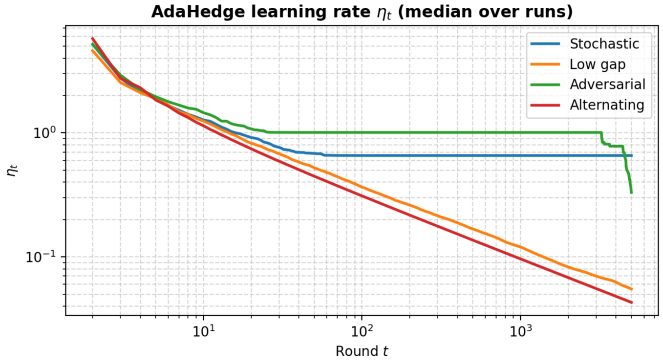


Figure 4: Median over runs of $\eta_t = \ln K/\Delta_{t-1}$ for AdaHedge ($t \geq 2$, log–log).

give the difference without a CI.

4.11 Verification of the regret–gap inequality

Table 4 checks Eq. (1) in every run: it holds in 20/20 runs in all settings. Moreover Δ_T is 3.6 on the easiest sequence and 53.6 on the hardest, always below $\sqrt{T \ln K} \approx 107$: the data-dependent quantity tracks the difficulty of the sequence, exactly as the learning-rate plot suggests. A weaker inequality, $R_T \leq \Delta_T + \ln K$, which I initially assumed, is *false* in general: it fails in 19/20 low-gap runs.

5 Discussion and limitations

When does AdaHedge help? Clearly when the best expert is identifiable (stochastic, switching): it removes the 7× (stochastic) and 4× (switching) overhead of the worst-case tuned η^*, and the scaling study shows this overhead grows with T. It protects against the failure of large η on deterministic hard data (6× lower regret than $\eta = 0.5$ at $T = 5000$, and the gap grows linearly with T). It gives no gain when the sequence is statistically too hard (low gap).

Negative aspects. (a) A hindsight-tuned Hedge is slightly better than AdaHedge on stochastic (4.2 vs. 5.5) and switching (8.3 vs. 9.6), by a small but statistically detectable margin; this oracle is not realisable and optimistic. (b) Nothing is distinguishable on low gap. (c) The "adversarial" sequence is oblivious and random; the alternating one is a single hand-built pattern. Adaptive

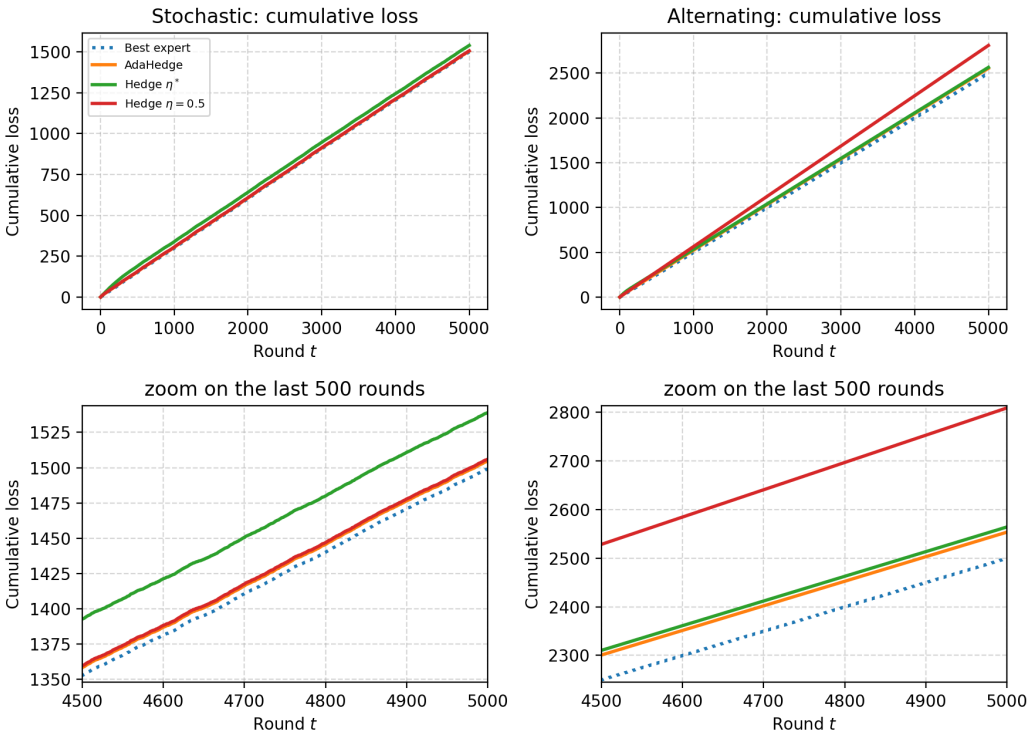


Figure 5: Mean cumulative loss (top) and a zoom on $t \in [4500, 5000]$ (bottom).

	vs. HEDGE η^*	vs. HEDGE $\eta = 0.5$	vs. oracle η
Stochastic	-34.4 ± 0.8	-1.3 ± 0.6	$+1.3 \pm 0.4$
Low gap	-0.2 ± 3.7	$+0.7 \pm 7.0$	$+2.1 \pm 4.9$
Adversarial	-28.1 ± 3.4	-2.5 ± 0.7	$+1.3 \pm 0.9$
Alternating	-10.6	-255.5	-9.6

Table 3: Paired difference in final regret, ADAHEDGE minus reference (mean ± 95% CI over 20 seeds). Negative: ADAHEDGE better.

adversaries were not tested, and neither HEDGE nor ADAHEDGE can track a switching comparator (Fixed Share would be needed).

Limitations. Only $K = 10$ and $T \leq 50\,000$; K was not varied; 20 seeds (10 at $T = 50\,000$); the η-grid is coarse and the oracle optimistic; losses are in $\{0, 1\}$ (the algorithms are defined for $[0, 1]$, fractional losses were not tested). The paper's FlipFlop algorithm, interpolating between HEDGE with large η and ADAHEDGE, was not implemented. The worst-case bound on Δ_T is quoted, not re-derived.

6 Conclusion

Fixed-η HEDGE forces a trade-off: large η is excellent on easy data and has linear regret on the worst case; the theory-optimal η^* is safe but slow, and its overhead on easy data grows with T. ADAHEDGE resolves this trade-off in my experiments: it beats every realisable fixed choice on the informative sequences, stays within a small margin of a hindsight-tuned HEDGE, and never needs T or the difficulty in advance. Its learning rate η_t and gap Δ_T give an interpretable measure of that difficulty, consistent with $R_T \leq 2\Delta_T$.

Setting	R_T	Δ_T	$2\Delta_T$	$R_T \leq 2\Delta_T$	$R_T \leq \Delta_T + \ln K$
Stochastic	5.5	3.6	7.3	20/20	13/20
Low gap	54.3	39.7	79.4	20/20	1/20
Switching	9.6	7.7	15.5	20/20	15/20
Alternating	54.1	53.6	107.2	20/20	20/20

Table 4: Mean R_T and Δ_T of AdaHedge at $T = 5000$, and the number of runs (of 20) in which each inequality holds. For reference $\sqrt{T \ln K} \approx 107$.

A Cumulative loss plots

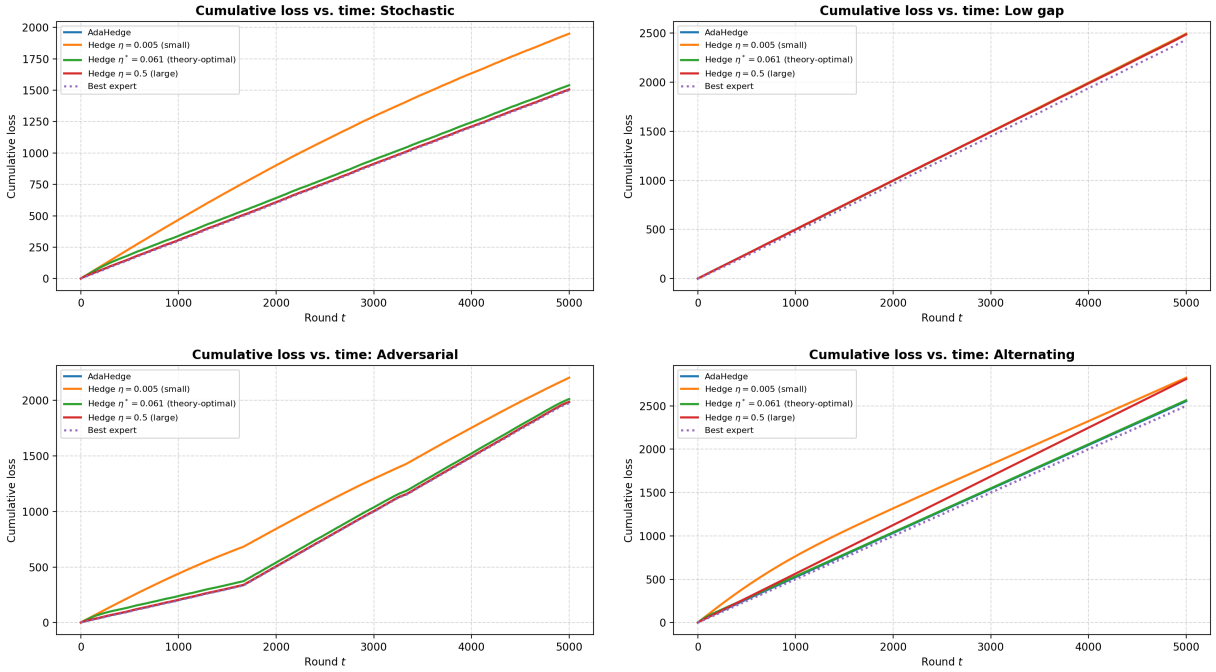


Figure 6: Cumulative loss vs. time (mean over 20 runs). At this scale, differences of a few tens of units are hard to see against totals of ≈ 2500; see Figure 5 for a zoom.

B Reproducibility

Code: `src/algorithms.py`, `src/datasets.py`, `main.py` (figures, `results/summary.csv`), `check_bound.py` (Table 4), `extra_experiments.py` (sweep with bound, scaling, paired statistics, pseudo-regret, zoomed plot) and `make_tables.py` (LaTeX tables generated from the CSV files, so report numbers match the code). Seeds are $0, \ldots, 19$ via `numpy.random.default_rng`. Run, from the repository root:

```
pip install -r requirements.txt
python main.py
python check_bound.py
python extra_experiments.py
python make_tables.py
latexmk -pdf report.tex
```

Python, NumPy and Matplotlib only; the runtime is a few minutes.

References

[1] T. van Erven, P. Grünwald, W. Koolen, S. de Rooij. *Adaptive Hedge*. NeurIPS 2011.

[2] S. de Rooij, T. van Erven, P. Grünwald, W. Koolen. *Follow the Leader If You Can, Hedge If You Must*. JMLR 15:1281–1316, 2014.